Notes: Ashcraft (AER, 2005)

Are Banks Really Special? New Evidence from the FDIC-Induced Failure of Healthy Banks

Contents

1 Big picture

- **Question.** Do bank failures still have real economic costs in the modern U.S. banking system, after deposit insurance, FDIC receivership, and a less bank-dependent economy have reduced many classic failure channels?
- **Core empirical challenge.** Ordinary bank failures are endogenous: banks fail when local borrowers and local economic conditions are already weak. A simple regression of county income on bank failures may confuse the effect of failure with pre-existing distress.
- **Natural experiment.** The paper studies two episodes in Texas in which subsidiaries that were relatively healthy failed because of their relationship to unhealthy lead banks in the same multi-bank holding company.
- **Main cases.** The first case is First City Bancorporation in 1992, where the FDIC used FIRREA's cross-guarantee provision. The second is First RepublicBank Corporation in 1988, where open-bank assistance to lead banks was not renewed and collateral claims swept other subsidiaries into failure.
- **Why the cases are useful.** In these episodes, failure was not primarily caused by local conditions in the markets served by the healthy subsidiaries. This helps isolate the effect of losing a bank relationship from the effect of weak borrowers.
- **Mechanism emphasized.** Because insured and uninsured creditors were largely protected, the main remaining channel is the destruction or disruption of lending relationships, especially information-intensive commercial lending.
- **Headline result.** Healthy-bank failures are followed by meaningful and persistent declines in local real county income. For a failure with deposits equal to 15% of county income, the estimated long-run decline is roughly 2–3%.
- **Economic takeaway.** Banks remain special when their lending relationships are difficult to replace. Small or relationship-intensive credit shocks may matter less in normal times, but the forced failure of a healthy bank can destroy valuable borrower-bank information and reduce local activity.

2 Incremental contribution of the paper

- **From correlation to quasi-experiment.** Earlier studies of bank failures often relied on correlations between failed deposits and local output. This paper improves identification by studying failures induced by FDIC actions that were plausibly unrelated to local county fundamentals.
- **Modern evidence on bank specialness.** The Great Depression literature suggests that bank failures were macroeconomically important. Ashcraft asks whether failures still matter under the modern institutional environment of deposit insurance and active resolution policy.
- **Focus on healthy-bank failures.** The paper turns an unusual regulatory event into an empirical design. Healthy subsidiaries are useful because their failure is less likely to reveal local economic weakness.

- **Mechanism decomposition.** By emphasizing that creditors were protected, the paper narrows the plausible channel to the loss of lending relationships rather than depositor wealth losses, liquidity freezes, or contagion.
- **Direct evidence on lending.** The paper uses pro forma bank balance sheets to show that commercial and industrial loans and unused commercial commitments contracted after healthy-bank failures.
- **Comparison with ordinary failures.** The paper first shows that average bank failures are followed by large income declines, but then explains why OLS estimates may overstate causal effects. The healthy-bank design helps discipline that interpretation.
- **Contribution to the lending-channel debate.** The paper helps reconcile why some bank loan supply shocks appear to affect output while others do not: firms may absorb small shocks, but not large disruptions to relationship lending.

3 Data and measurement

3.1 Failure events and banking data

- The paper uses the FDIC's *Historical Statistics on Banking* to identify commercial bank and thrift failures and assistance transactions from 1969 to 2002.
- Failed-deposit exposure is measured as failed-bank deposits divided by county income.
- The FDIC Summary of Deposits is used to map bank offices and deposits into counties.
- Call Reports provide bank-level balance sheet variables such as loans, liquid assets, problem loans, capital ratios, charge-offs, and return on assets.
- County-level bank variables are constructed by weighting bank characteristics by deposit market shares within each county.

Interpretation: The paper measures the intensity of a failure shock by the size of the failed institution relative to the local economy. This is more informative than a simple failure dummy because a small failed bank and a large failed bank should not have the same expected effect.

3.2 County income and local economic activity

- The main outcome is real county income from the Bureau of Economic Analysis.
- The panel covers U.S. counties from 1969 to 2000 in the OLS analysis.
- Real values are constructed by deflating county income with the national consumer price index.
- For the natural experiments, the unit of observation is a Texas county around the years 1988 and 1992.

Interpretation: County income is a broad measure of local economic activity. It is not a direct measure of borrower outcomes, but it allows the paper to ask whether banking shocks have real local consequences.

3.3 The First City Bancorporation experiment

- First City Bancorporation failed in October 1992 when regulators closed the Dallas and Houston lead banks.
- The FDIC expected losses of about $500 million from the lead banks.
- Under FIRREA's cross-guarantee authority, the FDIC charged expected losses against the capital of related solvent subsidiaries.
- Because the remaining subsidiaries had only $276 million in primary capital, the FDIC also closed these banks.
- Sixteen non-lead subsidiaries are treated as healthy-bank failures in the analysis; Austin and San Antonio are excluded from the healthy group because of weak capital positions.
- The resolution ultimately involved no loss to the FDIC and creditors with valid claims were eventually made whole.

Interpretation: The First City event is clean because the healthy subsidiaries did not fail because of their own poor loan books. Their failure was a legal and regulatory consequence of losses at related lead banks.

3.4 The First RepublicBank experiment

- First RepublicBank Corporation failed in July 1988 and was one of the largest and costliest failures in FDIC history at the time.
- Earlier in 1988, the FDIC provided $1 billion in open-bank assistance to the lead banks.
- The assistance note was guaranteed by the other subsidiaries and collateralized by the parent's equity in them.
- When the assistance was not renewed, the lead bank defaulted and the FDIC charged losses against the other subsidiaries.
- All forty subsidiary banks were placed into a single bridge bank, NCNB Texas.

Interpretation: The First RepublicBank case is less clean for lending measurement because healthy and problem subsidiaries were combined into one bridge bank. Still, the institutional reason for failure provides useful variation for studying the local real effects of relationship-bank disruption.

4 Models and empirical specifications

4.1 OLS first-pass specification

For the national county panel, the paper estimates

$$\ln(y_{c,t+k}) = \sum_{j=1}^{3} \beta_j \ln(y_{c,t-j}) + \delta_k \theta_{c,t} + \varepsilon_{c,t+k}, \quad k \in \{0, 1, \ldots, 6\},$$

where $y_{c,t+k}$ is real county income k years after the failure year, and $\theta_{c,t}$ is failed deposits divided by county income.

- Three lags of county income control for pre-existing local income levels and trends.
- Time effects are included in the regressions.
- Standard errors are robust and clustered at the county level.
- Separate coefficients are estimated for commercial bank failures, thrift failures, small versus large bank failures, and different resolution types.

Interpretation: This first-pass specification shows what a standard correlational analysis would imply. It is useful as a benchmark, but it cannot fully solve the endogeneity of ordinary bank failures.

4.2 Healthy-bank failure cross-section specification

For the natural experiments, the paper estimates a cross-sectional model across Texas counties:

$$\ln(y_{c,t+k}) = \alpha + \sum_{j=1}^{3} \beta_j \ln(y_{c,t-j}) + \sum_{j=1}^{3} \gamma_j x_{c,t-j} + \delta_k^H \theta_{c,t}^H + \delta_k^U \theta_{c,t}^U + \varepsilon_{c,t+k}.$$

- $\theta_{c,t}^H$ is healthy failed-bank deposits divided by county income.
- $\theta_{c,t}^U$ is unhealthy failed-bank deposits divided by county income.
- The control vector $x_{c,t-j}$ includes lagged county banking characteristics.
- Controls include county deposits, deposit Herfindahl index, failed-bank and failed-thrift market shares, deposit-weighted balance sheet variables, and in 1992, average exam ratings.
- The paper also estimates the same idea using changes in log income, then cumulates the coefficients across leads.

Interpretation: The coefficient δ_k^H is the key estimate. It asks whether counties exposed to the forced failure of healthy subsidiaries experienced lower future income, conditional on pre-failure local banking and income conditions.

4.3 Lending mechanism evidence

The paper also examines pro forma bank lending around failure:

- Commercial and industrial loans.
- Loans secured by real estate.
- Unused total loan commitments.
- Unused retail commitments.
- Unused commercial and industrial commitments.

Interpretation: Changes in loan stocks are imperfect measures of loan supply because failed-bank resolution can move problem assets across accounts. The unused commitment evidence is especially informative because it is less likely to be mechanically distorted by loan cleanup.

5 Identification and empirical design

5.1 What makes ordinary failures hard to interpret

- Banks usually fail because borrowers become distressed or local conditions deteriorate.
- If failures are caused by weak local fundamentals, then post-failure income declines may reflect the same underlying shock rather than the failure itself.
- Commercial bank failures may be especially endogenous because bank loan portfolios are tightly connected to local business conditions.
- Small bank failures may also be more endogenous because small banks lend more locally.

Interpretation: The paper's OLS results are consistent with banks being special, but they are not enough. The healthy-bank failures are used to address precisely this selection problem.

5.2 Why healthy-bank failures help

- Healthy subsidiaries failed because of holding-company obligations and FDIC resolution rules, not because local borrowers in those counties suddenly performed poorly.
- Figures 1A–1D show that healthy subsidiaries looked much more like nonfailing peer banks than like unhealthy failing banks before failure.
- Since depositors and creditors were largely protected, the failure shock does not primarily operate through depositor wealth loss or liquidity loss.
- The main remaining channel is the disruption of borrower-bank relationships.

Interpretation: The design is close to the ideal experiment: a bank lending relationship is removed for reasons plausibly outside the local economy. The question is whether borrowers can replace the bank on similar terms.

5.3 Remaining limitations

- The healthy subsidiaries were not randomly located across Texas counties.
- Affected counties were more urban, had higher deposits and income, and lower banking concentration.
- The paper addresses this by controlling for lagged county income, deposits, concentration, and bank balance sheet variables.
- Balance-sheet measures of lending after failure may still be affected by resolution accounting.

- The First RepublicBank lending evidence is less clean because all subsidiaries were placed into one bridge bank.

Interpretation: The identification is stronger than a standard bank-failure regression, but it is not a randomized experiment. The results should be read as strong quasi-experimental evidence that the destruction of healthy bank relationships has real effects.

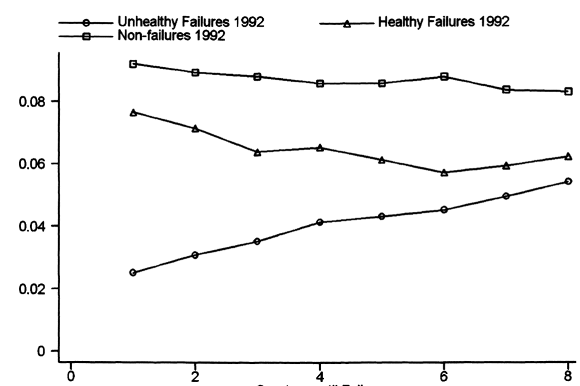
6 Tables and figures

6.1 Figures 1A and 1B: Healthy and unhealthy First City banks before failure

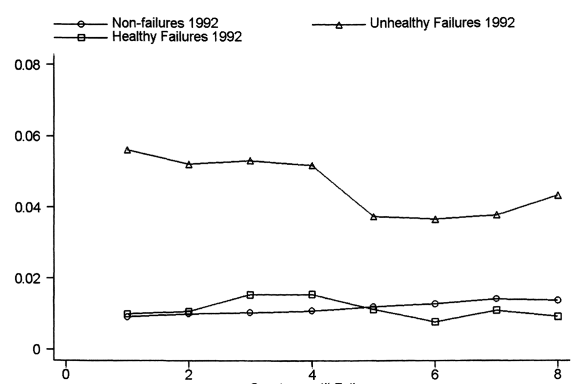
Read:

- Figure 1A compares primary capital-to-assets for 1992 Texas banks. Healthy First City subsidiaries had lower capital than nonfailing peers, but their capital ratios were far above those of unhealthy failures.
- Figure 1B compares nonperforming loans-to-assets in the same 1992 setting. Healthy First City subsidiaries looked much closer to nonfailures than to unhealthy failures.
- Together, the figures support the claim that healthy subsidiaries were not failing because their own local loan books resembled those of ordinary failing banks.

Economic message: The First City subsidiaries were not perfectly identical to peer banks, but they were much healthier than ordinary failed banks. This is the visual foundation for treating their failure as special and plausibly exogenous to local county conditions.



(a) Figure 1A: Primary capital to assets, 1992



(b) Figure 1B: Nonperforming loans to assets, 1992

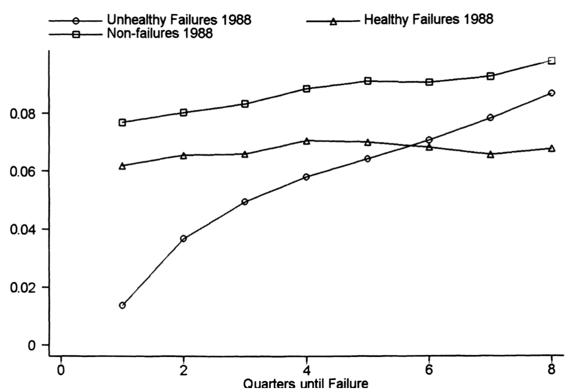
Figures 1A and 1B: First City Bancorporation subsidiaries before failure.

6.2 Figures 1C and 1D: Healthy and unhealthy First RepublicBank banks before failure

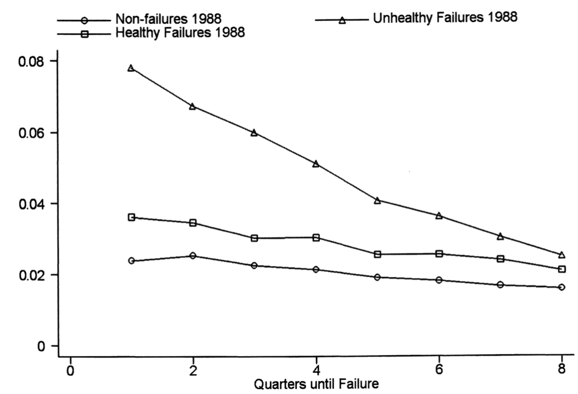
Read:

- Figure 1C compares primary capital-to-assets for 1988 Texas banks. Healthy First RepublicBank subsidiaries again track nonfailing peer banks more closely than unhealthy failures.
- Figure 1D shows nonperforming loans-to-assets. Healthy First RepublicBank subsidiaries had more problem loans than peers, but the level and trend are still far less distressed than unhealthy failures.
- The figures show why the paper calls these banks "healthy" in a relative sense, not because they had no problems at all.

Economic message: The First RepublicBank evidence is somewhat noisier than First City, but the basic comparison remains: the induced failures are distinct from ordinary unhealthy bank failures.



(a) Figure 1C: Primary capital to assets, 1988



(b) Figure 1D: Nonperforming loans to assets, 1988

Figures 1C and 1D: First RepublicBank subsidiaries before failure.

6.3 Tables 1 and 2: OLS benchmark and lending mechanism

Read:

- Table 1 reports OLS estimates for the effect of failed deposits on future real county income. Ordinary bank failures are followed by large and persistent income declines.
- For a bank failure with deposits equal to 15% of county income, Table 1 implies a 1.12% income decline in the failure year and about 2.83% after six years.
- Thrift failures have much smaller associations with income declines, consistent with banks being more information-intensive lenders than thrifts.
- Small-bank failures have much larger effects per dollar of deposits than large-bank failures.
- Table 2 examines lending around the two healthy-bank failures. For First City, commercial and industrial loans and unused commercial and industrial commitments decline sharply relative to peers.
- In First City, unused commercial and industrial commitments remain depressed, while retail commitments rebound more strongly.

Economic message: Table 1 motivates the paper by showing that bank failures are followed by local downturns, but Table 2 moves toward mechanism. The strongest evidence points to a contraction in information-intensive lending, not merely a mechanical reshuffling of deposits.

TABLE 1—OLS ESTIMATES OF THE EFFECT OF FAILURE ON REAL COUNTY INCOME

	Lead k of real county income						
	$k = 0$	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$	$k = 6$
Panel A. Ratio of failed deposits to income							
δ_k^{bank}	−0.0749***	−0.1030***	−0.1569***	−0.1774***	−0.1741***	−0.1941***	−0.1885***
	(0.0120)	(0.0175)	(0.0186)	(0.0228)	(0.0244)	(0.0267)	(0.0305)
Evaluated at $\theta_{c,t} = 0.15$	−1.12%	−1.55%	−2.35%	−2.66%	−2.61%	−2.91%	−2.83%
δ_k^{thrift}	−0.0060***	−0.0099**	−0.0074*	−0.0073**	−0.0097**	−0.0081	−0.0084
	(0.0020)	(0.0040)	(0.0044)	(0.0034)	(0.0053)	(0.0050)	(0.0061)
Evaluated at $\theta_{c,t} = 0.15$	−0.09%	−0.15%	−0.11%	−0.11%	−0.15%	−0.12%	−0.13%
Panel B. Ratio of failed deposits to income broken out across size and resolution type							
$\delta_k^{\text{small bank}}$	−0.1218***	−0.2085***	−0.2550***	−0.2310***	−0.2843***	−0.3519***	−0.3585***
	(0.0327)	(0.0414)	(0.0514)	(0.0617)	(0.0690)	(0.0798)	(0.0894)
Evaluated at $\theta_{c,t} = 0.15$	−1.83%	−3.13%	−3.83%	−3.47%	−4.26%	−5.28%	−5.38%
$\delta_k^{\text{large bank}}$	−0.0238	−0.0251	−0.0563	−0.0607	−0.0694	−0.0860	−0.0770
	(0.0289)	(0.0336)	(0.0418)	(0.0492)	(0.0536)	(0.0624)	(0.0694)
Evaluated at $\theta_{c,t} = 0.15$	−0.36%	−0.38%	−0.84%	−0.91%	−1.04%	−1.29%	−1.16%
$\delta_k^{\text{type II bank}}$	−0.0340	−0.0433	−0.0611	−0.0778	−0.0525	−0.0541	−0.0522
	(0.0307)	(0.0360)	(0.0450)	(0.0540)	(0.0583)	(0.0689)	(0.0769)
Evaluated at $\theta_{c,t} = 0.15$	−0.51%	−0.65%	−0.92%	−1.17%	−0.79%	−0.81%	−0.78%
$\delta_k^{\text{type III bank}}$	−0.0839	−0.1578**	−0.1583*	−0.2553**	−0.2750**	−0.1856	−0.2124
	(0.0580)	(0.0745)	(0.0882)	(0.1066)	(0.1237)	(0.1210)	(0.1401)
Evaluated at $\theta_{c,t} = 0.15$	−1.26%	−2.37%	−2.37%	−3.83%	−4.13%	−2.78%	−3.19%
Observations	88,798	85,724	82,650	79,576	76,502	73,428	70,356

Notes: The table reports coefficients and standard errors from OLS estimates of δ_k from equation (1) in the text: $\ln(y_{c,t+k}) = \sum_{j=1}^{3} \beta_j \ln(y_{c,t-j}) + \delta_k \theta_{c,t}$, where $\theta_{c,t}$ is equal to is the ratio of failed-bank deposits to county income. The data are a balanced panel of U.S. counties 1969–2000. In panel B, small failures are defined using the ninetieth percentile of the ratio of deposits to income. Type II failures refer to assisted mergers, while Type III failures refer to closures. Standard errors have been corrected for heteroskedasticity and clustered at the county level. Coefficients accented by one, two, and three asterisks are

(a) Table 1: OLS estimates of failure and county income

TABLE 2—BANK FAILURE, LOANS, AND LOAN COMMITMENTS

	First City Bancorporation (t = 1992)				First RepublicBank Corporation (t = 1988)			
Time	Healthy banks	Peer banks	Combined banks	Combined peer	All banks	Peer banks	Combined banks	Combined peer
Panel A. Commercial and industrial loans								
$t-2$	101.98%	115.38%	110.02%	114.29%	158.84%	135.86%	144.73%	124.03%
$t-1$	113.79%	98.99%	104.90%	105.54%	117.17%	119.33%	118.50%	109.73%
t	680,172	1,021,612	1,701,784	1,401,017	8,270,092	13,151,809	21,421,901	3,201,696
$t+1$	63.26%	98.11%	84.18%	102.03%	84.26%	100.55%	94.26%	90.42%
$t+2$	62.20%	101.84%	86.00%	105.40%	84.67%	94.85%	90.92%	78.33%
Panel B. Loans secured by real estate								
$t-2$	110.82%	94.73%	99.30%	100.75%	135.65%	109.36%	119.27%	93.31%
$t-1$	95.98%	95.53%	95.66%	98.11%	127.16%	108.06%	115.26%	98.71%
t	805,876	2,031,374	2,837,250	3,094,413	8,772,169	14,492,417	23,264,586	5,280,353
$t+1$	86.71%	108.75%	102.49%	101.57%	37.99%	95.34%	73.72%	97.73%
$t+2$	91.67%	119.63%	111.69%	105.82%	36.44%	89.63%	69.57%	94.24%
Panel C. Unused total loan commitments								
$t-2$	77.25%	79.72%	78.42%	88.20%	138.68%	130.92%	134.43%	90.95%
$t-1$	120.19%	92.42%	107.06%	97.94%	113.40%	129.94%	122.46%	90.94%
t	579,082	518,926	1,098,008	923,326	8,476,549	10,267,390	18,743,939	780,850
$t+1$	57.58%	129.99%	91.80%	113.30%	82.13%	111.33%	98.12%	107.10%
$t+2$	64.87%	163.12%	111.31%	125.58%	110.87%	130.10%	121.40%	138.25%
Panel D. Unused "retail" commitments								
$t-2$	86.19%	76.06%	78.05%	98.98%	N/A	N/A	N/A	N/A
$t-1$	84.70%	81.76%	82.33%	99.39%	N/A	N/A	N/A	N/A
t	35,506	145,891	181,397	126,501	N/A	N/A	N/A	N/A
$t+1$	73.64%	130.73%	119.56%	117.86%	N/A	N/A	N/A	N/A
$t+2$	230.51%	174.48%	185.45%	149.15%	N/A	N/A	N/A	N/A
Panel E. Unused "commercial & industrial" loan commitments								
$t-2$	76.67%	81.15%	78.49%	86.49%	N/A	N/A	N/A	N/A
$t-1$	122.51%	96.59%	111.96%	97.71%	N/A	N/A	N/A	N/A
t	543,576	373,035	916,611	796,825	N/A	N/A	N/A	N/A
$t+1$	56.53%	129.70%	86.31%	112.58%	N/A	N/A	N/A	N/A
$t+2$	54.05%	158.68%	96.63%	121.84%	N/A	N/A	N/A	N/A

Notes: The table displays June pro forma measures of lending. The first column refers to the pro forma institutions that acquired failed healthy banks; the second column refers to banks that had all deposits in the same counties as the acquiring banks in June 1994; the third column refers to the combination of the first two columns; and the fourth column refers to the pro forma banks that had no deposits in counties as institutions acquiring the failed First City banks. The last four columns are constructed in a similar fashion, but refer to all failed subsidiaries of First RepublicBank Corporation since they were placed into a single bridge bank.

(b) Table 2: Bank failure, loans, and commitments

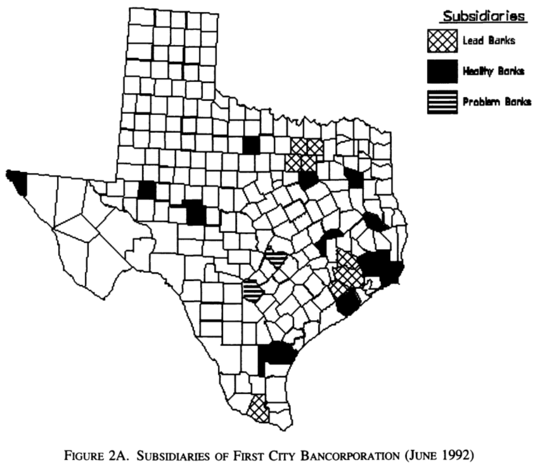
Tables 1 and 2: Benchmark correlations and lending mechanism.

6.4 Figures 2A and 2B: Geographic distribution of lead, healthy, and problem banks

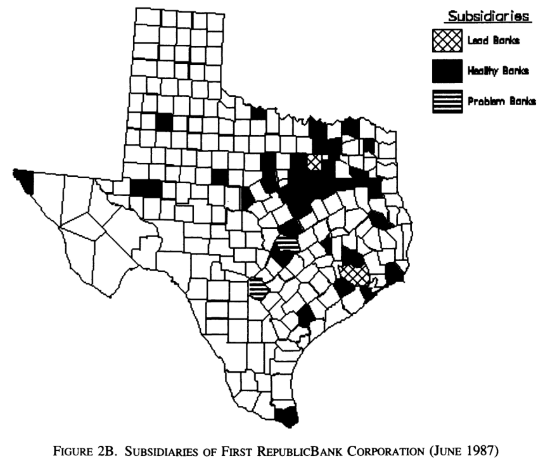
Read:

- Figure 2A maps First City Bancorporation subsidiaries in June 1992. Lead banks are concentrated in major metropolitan areas, while healthy-bank subsidiaries are spread across other Texas counties.
- Figure 2B maps First RepublicBank Corporation subsidiaries in June 1987. The healthy-bank exposure is geographically broad, with many affected counties outside the lead-bank locations.
- The maps show why a county-level empirical strategy is useful: the FDIC-induced failures created cross-county variation in exposure to healthy-bank failure.

Economic message: The treatment is not a single-city event. The holding-company structure spread the failure shock across many local banking markets in Texas.



(a) Figure 2A: First City subsidiaries, June 1992



(b) Figure 2B: First RepublicBank subsidiaries, June 1987

Figures 2A and 2B: Geographic exposure to induced failures.

6.5 Tables 3 and 4: Texas summary statistics and healthy-bank effects

Read:

- Table 3 compares affected and unaffected Texas counties. Affected counties are larger and more urban: they have higher lagged income and deposits and lower deposit Herfindahl indexes.
- The summary statistics show why controls are necessary. Healthy-bank exposure was not randomly assigned across counties.
- Table 4 presents the key natural-experiment estimates. In the First City 1992 cross section, a healthy-bank failure with deposits equal to 15% of income is associated with a 2.25% income decline after three years and 2.31% after six years.
- In the First RepublicBank 1988 cross section, the analogous healthy-bank failure effect is about 3.32% after six years.
- The estimates using changes in log income are broadly similar to those using levels.

Economic message: Table 4 is the core result. Even when failures are induced through the holding-company and FDIC resolution channel rather than local weakness, they are followed by persistent real declines. This is evidence that the destruction of bank relationships has macroeconomic value.

TABLE 3—TEXAS COUNTY SUMMARY STATISTICS

	First Republic (1988)		First City (1992)	
	Unaffected	Affected	Unaffected	Affected
$\theta_{c,t}^{healthy}$	0.0000	0.2166	0.0000	0.1834
	(0.0000)	(0.1523)	(0.0000)	(0.1841)
$\theta_{c,t}^{unhealthy}$	0.0328	0.1088	0.0053	0.0000
	(0.1014)	(0.1448)	(0.0370)	(0.0000)
$\text{Ln}(y_{c,t-1})$	12.24	14.35	12.51	14.23
	(1.27)	(1.04)	(1.44)	(1.04)
$\text{Ln}(\text{deposits}_{c,t-1})$	11.69	13.81	12.03	13.64
	(1.33)	(1.05)	(1.44)	(1.04)
$\theta_{c,t-1}^{unhealthy}$	0.0121	0.0161	0.0057	0.0000
	(0.0552)	(0.0405)	(0.0477)	(0.0000)
$\theta_{c,t-2}^{unhealthy}$	0.0181	0.0079	0.0178	0.0250
	(0.0908)	(0.0191)	(0.0584)	(0.0548)
Thrift failures $(t-1)$	0.0188	0.0571	0.0359	0.1176
	(0.1361)	(0.2355)	(0.1864)	(0.3321)
Thrift failures $(t-2)$	0.0657	0.0857	0.1570	0.1765
	(0.3153)	(0.2840)	(0.4909)	(0.3930)
Sensitivity to oil price	0.0028	0.0017	0.0026	0.0022
	(0.0043)	(0.0014)	(0.0042)	(0.0016)
$\text{HHI}_{c,t-1}$	4864	2152	4550	2392
	(2556)	(938)	(2461)	(1206)
Liquid assets $(t-1)$	0.5070	0.4610	0.5899	0.5758
	(0.1196)	(0.1095)	(0.1062)	(0.0710)
Primary capital ratio $(t-3)$	0.1017	0.0822	0.0987	0.0793
	(0.0226)	(0.0083)	(0.0269)	(0.0170)
Primary capital ratio $(t-2)$	0.1033	0.0792	0.0959	0.0791
	(0.0253)	(0.0090)	(0.0268)	(0.0158)
Primary capital ratio $(t-1)$	0.1002	0.0779	0.0970	0.0796
	(0.0256)	(0.0106)	(0.0274)	(0.0155)
Problem loans $(t-3)$	0.0129	0.0152	0.0160	0.0151
	(0.0101)	(0.0062)	(0.0143)	(0.0099)
Problem loans $(t-2)$	0.0165	0.0218	0.0136	0.0121
	(0.0116)	(0.0106)	(0.0121)	(0.0067)
Problem loans $(t-1)$	0.0181	0.0248	0.0114	0.0124
	(0.0157)	(0.0164)	(0.0103)	(0.0048)
Return on assets $(t-3)$	0.0094	0.0066	0.0042	0.0036
	(0.0075)	(0.0046)	(0.0108)	(0.0065)
Return on assets $(t-2)$	0.0061	−0.0004	0.0060	0.0051
	(0.0080)	(0.0073)	(0.0068)	(0.0045)
Return on assets $(t-1)$	0.0029	−0.0029	0.0076	0.0078
	(0.0108)	(0.0093)	(0.0064)	(0.0065)
	213	35	223	17

Notes: The table reports the mean and standard deviation for several county-level variables. The first two columns refer to Texas counties in 1988, while the second two columns refer to Texas counties in 1992. In each unaffected column, there were no healthy-bank failures, while in each affected column there was at least one healthy-bank failure. Note that $\theta_{c,t}^{x}$ refers to the ratio of healthy or unhealthy failed-bank deposits to income.

(a) Table 3: Texas county summary statistics

TABLE 4—THE EFFECT OF HEALTHY-BANK FAILURE ON REAL ACTIVITY

	Lead of real county income						
	$k = 0$	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 4$	$k = 6$
Panel A. First City Bancorporation (cross section of 240 Texas counties in 1992)							
Dependent variable: $ln(y_{t+k})$							
$\delta_k^{healthy}$	−0.0374	−0.0797*	−0.1315**	−0.1497**	−0.1378	−0.1347	−0.1537
	(0.0293)	(0.0407)	(0.0555)	(0.0736)	(0.1015)	(0.1070)	(0.1149)
Evaluated at $\theta_{c,t} = 0.15$	−0.0056	−0.0120	−0.0197	−0.0225	−0.0207	−0.0202	−0.0231
$\delta_k^{unhealthy}$	−0.1792***	−0.2716***	−0.2198**	−0.3942***	−0.6538***	−0.5023***	−0.5384***
	(0.0435)	(0.0744)	(0.1039)	(0.1239)	(0.1564)	(0.1318)	(0.1604)
Evaluated at $\theta_{c,t} = 0.15$	−0.0269	−0.0407	−0.0330	−0.0591	−0.0981	−0.0753	−0.0808
Dependent variable: $\Delta ln(y_{t+k})$							
$\delta_{k,c}^{healthy}$	−0.0400	−0.0554*	−0.0538*	−0.0210	−0.0012	0.0122	−0.0055
	(0.0295)	(0.0298)	(0.0316)	(0.0344)	(0.0402)	(0.0266)	(0.0296)
Sum of coefficients	−0.0400	−0.0954	−0.1492	−0.1702	−0.1690	−0.1568	−0.1623
Evaluated at $\theta_{c,t} = 0.15$	−0.0060	−0.0143	−0.0224	−0.0255	−0.0254	−0.0235	−0.0243
$\delta_k^{unhealthy}$	−0.1787***	−0.0923*	0.0511	−0.1753**	−0.2600***	0.1511**	−0.0384
	(0.0435)	(0.0538)	(0.0672)	(0.0778)	(0.0695)	(0.0598)	(0.0690)
Sum of coefficients	−0.1787	−0.2710	−0.2199	−0.3952	−0.6552	−0.5041	−0.5425
Evaluated at $\theta = 0.15$	−0.0268	−0.0407	−0.0330	−0.0593	−0.0983	−0.0756	−0.0814
Panel B. First RepublicBank Corporation (cross section of 248 Texas counties in 1988)							
Dependent variable: $ln(y_{t+k})$							
$\delta_{k,c}^{healthy}$	0.0029	−0.0564	−0.0766	−0.1299*	−0.1581**	−0.1912**	−0.2216**
	(0.0308)	(0.0442)	(0.0515)	(0.0671)	(0.0723)	(0.0820)	(0.0890)
Evaluated at $\theta_{c,t} = 0.15$	0.0004	−0.0085	−0.0115	−0.0195	−0.0237	−0.0287	−0.0332
$\delta_{k,c}^{unhealthy}$	0.0108	−0.0479	−0.0907**	−0.1016**	−0.1254**	−0.1566**	−0.1696**
	(0.0193)	(0.0318)	(0.0364)	(0.0482)	(0.0524)	(0.0619)	(0.0746)
Evaluated at $\theta_{c,t} = 0.15$	0.0016	−0.0072	−0.0136	−0.0152	−0.0188	−0.0235	−0.0254
Dependent variable: $\Delta ln(y_{t+k})$							
$\delta_k^{healthy}$	0.0029	−0.0597*	−0.0201	−0.0522	−0.0289	−0.0316	−0.0323
	(0.0309)	(0.0361)	(0.0297)	(0.0370)	(0.0247)	(0.0230)	(0.0304)
Sum of coefficients	0.0029	−0.0568	−0.0769	−0.1291	−0.158	−0.1896	−0.2219
Evaluated at $\theta_{c,t} = 0.15$	0.0004	−0.0085	−0.0115	−0.0194	−0.0237	−0.0284	−0.0333
$\delta_k^{unhealthy}$	0.0126	−0.0577**	−0.0478**	−0.0048	−0.0244	−0.0281	−0.0111
	(0.0195)	(0.0265)	(0.0209)	(0.0218)	(0.0159)	(0.0224)	(0.0246)
Sum of coefficients	0.0126	−0.0451	−0.0929	−0.0977	−0.1221	−0.1502	−0.1613
Evaluated at $\theta_{c,t} = 0.15$	0.0019	−0.0068	−0.0139	−0.0147	−0.0183	−0.0225	−0.0242

Notes: The table reports the coefficient estimate and standard error on $\delta_k^{healthy}$ and $\delta_k^{unhealthy}$ from estimation of equation (2) in the text:

$$Z_{c,t+k} = \alpha + \sum_{i=1}^{3} \beta_i Z_{c,t-j} + \sum_{i=1}^{3} \gamma_i x_{c,f-j} + \delta_k^{healthy}\theta_{c,t}^{healthy} + \delta_k^{healthy}\theta_{c,t}^{unhealthy} + \varepsilon_{c,t}$$

where $\theta_{c,t}^{healthy}$ and $\theta_{c,t}^{unhealthy}$ are the ratios of healthy and unhealthy failed-bank deposits to county income, respectively. In column k, the dependent variable $Z_{c,t+k}$ is either the level or change in the log of real county income k years after failure. Standard errors have been corrected for heteroskedasticity. Coefficients accented by one, two, and three asterisks are statistically significant at the 10-, 5-, and 1-percent levels, respectively.

(b) Table 4: Effect of healthy-bank failure on real activity

Tables 3 and 4: County balance and main natural-experiment estimates.

7 Short summary

- The paper asks whether banks are still special in the modern U.S. economy.
- Ordinary bank failures are followed by large and persistent county-income declines, but these OLS results are vulnerable to endogeneity.
- The paper studies two FDIC-induced failures of relatively healthy subsidiaries: First City Bancorporation in 1992 and First RepublicBank Corporation in 1988.
- These failures were caused by cross-guarantees or collateral claims related to distressed lead banks, not primarily by the local conditions facing the healthy subsidiaries.
- Depositors and creditors were largely protected, so the key channel is the disruption of lending relationships.
- Lending evidence shows sharp contractions in commercial and industrial lending and unused commercial commitments, especially for First City.

- County-level estimates show that healthy-bank failures led to persistent real income declines of roughly 2–3% for a failure equal to 15% of county income.
- The evidence supports the view that bank-specific relationships are economically valuable and can have local macroeconomic effects when destroyed.

8 Conclusion

8.1 Core conclusion

Ashcraft shows that banks remain special because they create and maintain lending relationships that are not instantly replaceable. When a healthy bank is forced to fail for reasons unrelated to local borrowers, the affected counties experience a meaningful and persistent decline in real income.

8.2 Economic intuition

The intuition is that many borrowers, especially small and relationship-dependent firms, cannot immediately substitute away from their existing bank. A bank may possess private information about borrower quality, collateral, local markets, and repayment histories. When the bank fails, that information and the associated credit lines may be destroyed or interrupted. Other lenders may be unwilling to extend credit on the same terms, causing investment, employment, or spending to fall.

8.3 Incremental contribution revisited

The paper's distinctive contribution is not simply to show that failed banks and weak economies move together. It identifies unusual cases where otherwise healthy banks failed because of regulatory and holding-company mechanisms. This turns bank failure into a cleaner shock to relationship lending and provides new evidence that banks matter beyond the Great Depression setting.

8.4 Implications for bank regulation

The findings help explain why bank failures receive special regulatory attention. Even when deposit insurance limits depositor losses and FDIC receivership limits liquidity disruption, the loss of bank-borrower relationships can still impose real costs on local economies. Resolution policy therefore matters not only for protecting depositors and taxpayers, but also for preserving lending relationships when feasible.

8.5 Limitations

The events are unusual and occur in Texas during a period of severe banking-sector stress. The healthy subsidiaries were not randomly located, and the sample contains only two major natural experiments. Balance-sheet lending evidence is also imperfect because bank resolution can reclassify assets. These limitations do not eliminate the paper's contribution, but they mean the results should be interpreted as evidence from a rare and informative setting rather than as a universal average treatment effect of all bank failures.

8.6 Takeaway

The main takeaway is that the macroeconomic importance of banks depends on the size and substitutability of the credit relationship shock. Small loan-supply shocks may be absorbed through other credit sources, but large disruptions to relationship banking can reduce local real activity. In that sense, banks are still special.